Joseph A. DePaolo-Boisvert

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EDUCATION

Doctor of Philosophy, Chemistry — Illinois Institute of Technology, Chicago, IL **Conferral August 2026**
Successfully defended April 2026. Advisor: Dr. David D. L. Minh. Master's-level requirements of the Ph.D. completed May 2022.
Dissertation: "Membrane Protein Molecular Dynamics: Application, Representation, Enhanced Sampling."

Bachelor of Arts, Chemistry; Bachelor of Arts, Physics — Boston University, Boston, MA **May 2020**
Coursework additionally completed at Worcester Polytechnic Institute and Trinity College. Two-year course series in Russian Language and Literature.

APPOINTMENTS

Co-Founder & Chief Technology Officer — Biagon Inc., Chicago, IL **Feb 2024 – Present**

Doctoral Researcher — Illinois Institute of Technology, Department of Chemistry **Aug 2020 – Aug 2026**
Minh Laboratory (Chemistry) and Juárez Laboratory (Biological Sciences).

Instructor of Record — CHEM 467 Medicinal Chemistry, Illinois Institute of Technology **Fall 2025**

Teaching Assistant — Illinois Institute of Technology, Department of Chemistry **Aug 2020 – Aug 2026**

First Mate — City Experiences by Hornblower, Chicago, IL **May 2024 – Present**
Senior Deckhand by United States Coast Guard designation. Spirit of Navy Pier.

PEER-REVIEWED PUBLICATIONS

1. **DePaolo-Boisvert, J. A.**; Tuz, K.; Minh, D. D. L.; Juárez, O. X. "Molecular Dynamics Analysis of Inhibitor Binding Interactions in the *Vibrio cholerae* Respiratory Complex NQR." *Proteins: Structure, Function, and Bioinformatics* **2025**, 94 (2), 649–659. doi:10.1002/prot.70036
2. Cooper, D. A.; **DePaolo-Boisvert, J.**; Nicholson, S. A.; Gad, B.; Minh, D. D. L. "Intracellular Pocket Conformations Determine Signaling Efficacy through the μ Opioid Receptor." *Journal of Chemical Information and Modeling* **2025**, 65 (3), 1465–1475. doi:10.1021/acs.jcim.4c01437
3. Sorescu, J. M.; González-Montalvo, M. A.; Yuan, M.; **DePaolo-Boisvert, J.**; Ceapă, C. D.; Garcia-Contreras, R.; Flores-Herrera, O.; Shea, M. E.; Tuz, K.; Juárez, O. X. "Breakthroughs in the Development of Antibiotics, Antifungals and Antiparasitics Targeting the Pathogens' Respiratory Chain." *Critical Reviews in Biochemistry and Molecular Biology* **2025**, 60 (4–6), 141–174. doi:10.1080/10409238.2025.2545785
4. González-Montalvo, M. A.; Sorescu, J. M.; Yuan, M.; **DePaolo-Boisvert, J.**; Liang, P.; Juárez, O. X.; Tuz, K. "NQR as a Target for New Antibiotics." *Frontiers in Microbiology* **2025**, 16, 1690572. doi:10.3389/fmicb.2025.1690572
5. Tuz, K.; Yuan, M.; Hu, Y.; Do, T. T. T.; Willow, S. Y.; **DePaolo-Boisvert, J. A.**; Fuller, J. R.; Minh, D. D. L.; Juárez, O. "Identification of the Riboflavin Cofactor-Binding Site in the *Vibrio cholerae* Ion-Pumping NQR Complex: A Novel Structural Motif in Redox Enzymes." *Journal of Biological Chemistry* **2022**, 298 (8), 102182. doi:10.1016/j.jbc.2022.102182

MANUSCRIPTS IN PREPARATION

6. **DePaolo-Boisvert, J. A.**; Willow, S. Y.; Srinarmwong, T.; Bahl, A.; Minh, D. D. L. "Deep Neural Network Auto-Encoding Models Reveal Inherent Dimensionality of Protein Structural Ensembles."

SOFTWARE

ChiMPSS — Chicago Membrane Protein Simulation Suite. Open-source infrastructure for reproducible enhanced-sampling molecular dynamics of membrane protein systems; three coordinated repositories at github.com/CCBatIIT.

- **MotorRow** — multi-stage equilibration burn-in. DePaolo-Boisvert, J. A.; Cooper, D. A.; Xie, B.; Nguyen, T.-V.-A. (2025). *First author*.
- **Bridgeport** — automated system preparation: structure repair, protonation, membrane embedding, ligand parameterization. Cooper, D. A.; DePaolo-Boisvert, J. A.; Nguyen, T.-V.-A. (2025).
- **FultonMarket** — adaptive parallel-tempering replica exchange with dynamic state insertion. Cooper, D. A.; DePaolo-Boisvert, J. A.; Nguyen, T.-V.-A.; Khan, A. (2025).

RESEARCH EXPERIENCE

Virtual Screening and Structure-Based Antibacterial Discovery — Na^+-Pumping NADH:Quinone Oxidoreductase

- Designed and executed a 4.2-million-compound virtual screen of the Enamine library against the NQR ubiquinone binding site on the Bridges2 supercomputer, partitioning the library into 50,000-compound batches distributed across full CPU nodes with one AutoDock Vina docking job per core.
- Advanced top-scoring hits through successive enumeration cycles, promoting leading compounds to scaffolds for focused libraries with systematic substitutions; surfaced pharmacophore hypotheses taken up by an inter-institutional medicinal chemistry collaboration (Becker, Loyola University; Juárez and Tuz, Illinois Tech).

- Produced the first molecular dynamics simulations of *Vibrio cholerae* NQR in complex with the substrate analogue ubiquinone-4 and three potent inhibitors (HQNO, aurachin-D42, korormicin-A): 13 production simulations of approximately 350–400k atoms each, run in triplicate for at least 500 ns in a heterogeneous POPE/POPG/cardiolipin membrane.
- Identified conserved hydrogen-bonding and hydrophobic binding motifs in subunit B through protein–ligand interaction fingerprinting, distance analysis, and binding pose clustering; interpreted results against published mutagenesis to establish the structure–activity basis for rational inhibitor design.
- Contributed docking and molecular dynamics analysis to an FDA-approved drug repurposing effort against NQR with the medicinal chemistry group of Dr. Daniel Becker, Loyola University.

Reproducible Enhanced Sampling of Class-A G Protein-Coupled Receptors

- Co-developed ChiMPSS and implemented parallel-tempering replica exchange with dynamic temperature-state insertion driven by inter-state mixing probability, MBAR equilibration detection, and importance resampling (pyMBAR).
- Validated the pipeline across four class-A GPCRs — cannabinoid receptor 2, serotonin receptors 2A and 2B, and the kappa opioid receptor — with triplicate microsecond-scale ensembles converging on consistent transmembrane helix conformational distributions from both common and structurally distinct starting configurations.
- Constructed a pan-GPCR reference frame by principal component analysis of 555 experimentally determined structures from GPCRdb; identified transducer-stabilization bias in crystallographic reference sets as a systematic confound for pharmacological inference from ensemble position.

Machine Learning Representations of Molecular Conformational Ensembles

- Developed stochastic autoencoder architectures in JAX and Flax for nonlinear dimensionality reduction of molecular dynamics trajectories; benchmarked systematically across five systems of increasing complexity — oxycodone, deca-alanine, crambin, a bromodomain with JQ1, and HIV-1 protease with a peptide ligand — against principal component analysis and deterministic autoencoding.
- Demonstrated that stochastic models resolve more distinct conformational clusters at low latent dimensionality and generalize better than deterministic autoencoding at the highest compression ratios; characterized intrinsic compression thresholds using nearest-neighbor intrinsic dimensionality estimation, HDBSCAN, and UMAP.
- Contributed simulation and analysis to machine learning models relating μ opioid receptor intracellular pocket conformations to G protein and β-arrestin-2 signaling efficacy, computing efficacy along multiple pathways rather than binary activity classification.

Polarization Corrections to Binding Free Energy

- Quantified protein-induced polarization of ligands within binding pockets and incorporated the resulting polarization energy as a corrective term in binding free energy calculations. Advisors: Dr. David D. L. Minh and Dr. Soohaeng Yoo Willow.

RESEARCH SUPPORT AND COMPUTING ALLOCATIONS

- **Graduate Principal Investigator** — ACCESS allocation CIS250014. Principal manager of an inherited XSEDE-era allocation (MCB150144).
- Authored allocation applications on behalf of Dr. Oscar Juárez and renewal reports on behalf of Dr. David Minh across nine ACCESS allocations supporting the dissertation work: MCB150144, BIO230006, BIO240146, BIO240217, CIS250014, BIO250102, BIO250114, BIO250326, BIO260063. Applications subsequently adopted as templates by other students securing independent awards.
- Tens of thousands of GPU-hours consumed across Bridges2 (Pittsburgh Supercomputing Center), Expanse (San Diego Supercomputing Center), and Jetstream2 (Indiana University).
- Dissertation research supported by NIH R01AI151152 (Juárez and Tuz), NIH R01GM127712 (Minh), NSF Award #2331458, and Walder Foundation Grant 23-01215.

TEACHING AND MENTORSHIP

Instructor of Record — CHEM 467, Medicinal Chemistry — Illinois Institute of Technology **Fall 2025**

- Sole instructor for an upper-division medicinal chemistry lecture course; held full responsibility for curriculum, instruction, assessment, and grading across a twenty-eight-session semester.
- Designed the syllabus in three modules: (i) introduction to biochemistry and biological macromolecules as drug targets — protein structure, ligand–protein complementarity, enzymes and receptors as targets, binding kinetics, oligonucleotides as targets; (ii) properties and reactions of drugs in the human body — pharmacokinetics and metabolism; (iii) drug discovery, optimization, and regulation — molecular structure and diversity, lead discovery, and lead optimization. Texts: Stevens, *Medicinal Chemistry: The Modern Drug Discovery Process*; Nelson and Cox, *Lehninger Principles of Biochemistry*.
- Curated a fourteen-paper primary literature reading list paired to lecture topics, ranging from the acetylation of prostaglandin synthase by aspirin to AutoDock Vina 1.2.0 and R-group replacement databases for medicinal chemistry.
- Set and graded all assessment: two non-cumulative midterm examinations, weekly problem sets, two student journal-club presentations per student, and a capstone report and presentation analyzing the discovery, mechanism of action, target interactions, pharmacokinetics, and clearance of a student-selected FDA-approved drug.
- Recruited and hosted external guest lecturers from academic and industry practice, including Dr. Oscar Juárez (Illinois Tech, Biological Sciences) and the Chief Executive Officer of Biagon Inc.

Teaching Assistant — Illinois Institute of Technology **Aug 2020 – Aug 2026**

- Six semesters of instruction, Undergraduate Physical Chemistry Laboratory (Dr. Benjamin Zion).
- Two semesters of grading, Undergraduate Biochemistry (Dr. Oscar Juárez).

Instructor — Molecular Simulation and Calculation Workshop, Barranquilla, Colombia **Mar 2022**

- Taught binding free energy calculations and force field parameterization to an international cohort of researchers.

Research Mentorship **Ongoing**

- Onboarded three graduate students onto the ChiMPSS codebase; the suite is now the standard pipeline of the Minh group across six receptor targets.
- Taught and helped train a physical chemistry laboratory student who moved into undergraduate research in the Minh group; now a shared co-author, ChiMPSS co-developer, and Chief Executive Officer of Biagon Inc.

PRESENTATIONS

Oral — "Energy-Preserving Variational Auto-Encoding and Decoding of Atomistic Protein Structure." American Chemical Society National Meeting, San Francisco, August 2023; Gibbs 37 Conference on Biothermodynamics, October 2023.

Poster — "Solvation Free Energy Calculations using RealNVP Normalizing Flows." American Chemical Society National Meeting, Chicago, August 2022; Gibbs 36 Conference on Biothermodynamics, October 2022.

Poster — Annual graduate research presentations, Illinois Institute of Technology, 2021–2025 (topics vary).

PROFESSIONAL DEVELOPMENT

- **Babson College SUD Sprint** — 2025 cohort member, Babson Healthcare Entrepreneurship Center.
- **Nucleate Startup Accelerator** (Spring 2025); **IIT Kaplan Institute Startup Accelerator** (Spring 2024); **Loyola I-Corps Program** (Fall 2023).
- **BIO International Convention** — attended annually since 2024; business development with contract research organizations and prospective partners on behalf of Biagon Inc.
- **ACS–NVIDIA Machine Learning Workshop** — American Chemical Society National Meeting, Chicago, August 2022. Awarded the ACS–NVIDIA Certificate of Machine Learning Knowledge.

TECHNICAL SKILLS

Computer-Aided Drug Design: Molecular docking and virtual screening at multi-million-compound scale (AutoDock Vina, MGLTools); iterative hit expansion and focused library enumeration; pharmacophore identification; protein–ligand interaction fingerprinting (ProLIF); binding pose clustering; alchemical and machine-learning binding free energy calculation (YANK); ligand polarization corrections; structure-guided structure–activity relationship development.

Molecular Dynamics and Enhanced Sampling: OpenMM; parallel-tempering replica exchange with adaptive state insertion; MBAR equilibration detection and importance resampling (pyMBAR); membrane protein system construction (CHARMM-GUI, PDBFixer); structure repair and homology modeling (MODELLER, AlphaFold); protonation (PDB2PQR, AmberTools); force field parameterization and concatenation across ff14SB, Lipid17, OPC3/OPC, GAFF2, and OpenFF/SMIRNOFF, with AM1-BCC charge assignment via antechamber.

Machine Learning: JAX, Flax; stochastic and deterministic autoencoders; normalizing flows (RealNVP); nonlinear dimensionality reduction and intrinsic dimensionality estimation; HDBSCAN, UMAP, principal component analysis, scikit-learn.

Cheminformatics and Data: RDKit (conformer generation, maximal common substructure alignment, protonation); OpenFF Toolkit; SMILES handling; mining and curation from the Protein Data Bank, PubChem, GPCRdb, OPM, and Enamine libraries; MDTraj, MDAnalysis, NumPy, SciPy, Pandas, Matplotlib, Jupyter, NetCDF.

High-Performance Computing: Bridges2, Expanse, and Jetstream2; SLURM, PBS, and VM-based environments; CPU-parallel docking campaigns via batch partitioning and multiprocessing; GPU-accelerated molecular dynamics.

Programming: Python (expert), Bash, C; Git and GitHub; Linux.

CERTIFICATIONS AND LICENSES

- ACS–NVIDIA Certificate of Machine Learning Knowledge (2022).
- Senior Deckhand, United States Coast Guard designation (2024).
- 100-Ton Master of Motor Vessels, Great Lakes and Inland, and OUPV Near Coastal, Great Lakes and Inland — application submitted and under United States Coast Guard review.

LANGUAGES

English (native). Russian (reading proficiency; two-year university course series in language and literature).